

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – ConstraintBased Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	CSE-AI	Date:	26-08-2026

Code of Conduct

I N Goutham/192472024(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:N Goutham

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability) ■ Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. **Conversation History:** User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse “exam”, “concentrate”, “you”).
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. **Source Sentence:** “The bank by the river flooded after the storm, but it was saved by quick action.”

Note: “bank” is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

- b) Write a predicate logic representation of the sentence.
- c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.
- d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.
- e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

Q1. Reference Resolution and Text Coherence

Given paragraph:

“John and Mary went to the park. He brought a ball.

She wanted to play with it. The dog chased him

excitedly. Finally, they all went home.”

The assessment asks us to identify referring expressions, apply gender/number, recency, semantic compatibility and coherence constraints, and produce the final coreference chains.

a) Referring Expressions and Possible Antecedents

Referring Expression	Possible Antecedents	Final Antecedent
He	John, Mary	John
She	John, Mary	Mary
it	ball, park	ball
him	John, Mary	John
they	John, Mary, dog	John + Mary + dog
all	John, Mary, dog	John + Mary + dog

b) Applying Constraints

1. Gender and Number Agreement

- He → John because John is male and singular.
- She → Mary because Mary is female and singular.
- it → ball because “ball” is singular and non-human.
- him → John because John is male and singular.
- they → John + Mary + dog because the pronoun is plural.
- all → John + Mary + dog because “all” refers to the complete group.

2. Recency Constraint

The most recently mentioned compatible entity is preferred.

For example:

- “She wanted to play with **it**.”
- The nearest suitable antecedent for “it” is **ball**, rather than park.

3. Semantic Compatibility

The antecedent must make sense with the action.

- “Mary wanted to play with **the ball**” is semantically appropriate.
- “The dog chased **John**” is a meaningful action.
- “John, Mary and the dog went home” is coherent.

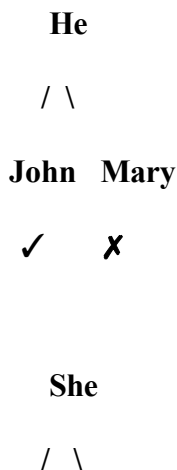
4. Coherence Constraint

The same entity must maintain a consistent reference throughout the discourse.

c) Constraint Table

Pronoun	Initial Domain	Constraints Applied	Eliminated	Selected
He	John, Mary	Gender + number	Mary	John
She	John, Mary	Gender + number	John	Mary
it	ball, park	Semantics + recency	park	Ball
him	John, Mary	Gender + semantics	Mary	John
they	John, Mary, dog	Number + discourse	Singular entities	John + Mary + dog
all	John, Mary, dog	Number + coherence	Individual entities	John + Mary + dog

Constraint graph:



John Mary

✗ ✓

it

**/ **

ball park

✓ ✗

him

**/ **

John Mary

✓ ✗

they

**/ | **

John Mary Dog

\ | /

✓ GROUP ✓

d) Final Coreference Chains

John → He → him → they → all

Mary → She → they → all

ball → it

dog → they → all

Rewritten Paragraph

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

e) Priority of Constraints

The recommended priority is:

1. Gender and number agreement
2. Semantic compatibility

3. Recency
4. Coherence

Agreement is applied first because an antecedent violating grammatical agreement should normally be rejected. Semantic compatibility then removes references that do not make sense.

Recency helps select among remaining candidates, while coherence ensures that the complete discourse maintains consistent entity chains.

If the recency constraint is relaxed, the system can still resolve references using gender, number, semantics and discourse coherence. However, when multiple candidates are grammatically possible, the system may have difficulty selecting the intended antecedent.

Q2. Constraint-Based Dialog Generation

The conversation is:

User: “I have an important exam tomorrow but I’m not able to concentrate.”

The required dialog act is **Advise + Encourage**, with specific constraints on entity coherence, discourse relation, keywords, length and tone.

a) Discourse Planning Steps

The following steps are required:

1. Identify the user's main problem — difficulty concentrating.
2. Identify the context — an important exam tomorrow.
3. Determine the dialog acts — **Advice + Encourage**.
4. Select a discourse relation — **Cause-Effect** or **Elaboration**.
5. Include at least two keywords — **focus, break, confident**.
6. Generate a polite and positive response.
7. Verify that it contains exactly 2–3 sentences.
8. Check logical consistency.

b) Three Possible Responses

Response 1:

“Since your exam is tomorrow, take a short break and return with a clear focus on the most important topics. Stay confident and remind yourself that steady preparation can help you perform well.”

Response 2:

“Your exam is important, so try studying in short sessions with small breaks to improve your

focus. You can stay confident by concentrating on one topic at a time.”

Response 3:

“Because your exam is tomorrow, take a short break and then focus on one topic at a time. Stay confident and avoid worrying about everything at once.”

Q3. Word Sense Disambiguation and Predicate Logic

The source sentence is:

“The bank by the river flooded after the storm, but it was saved by quick action.”

The assessment specifically identifies **bank** as ambiguous and requires WSD, predicate logic, discourse structure and a coherent paraphrase.

a) Word Sense Disambiguation

The word “**bank**” has two possible meanings:

1. Financial institution
2. Riverbank / land beside a river

The correct sense is:

Bank = riverbank

Justification

The phrase “by the river” strongly indicates that the word refers to the land beside a river. Also, a riverbank can naturally flood, whereas a financial institution does not normally flood in the literal sense described here.

Therefore:

bank

|—— Financial institution ✗

└—— Riverbank ✓

b) Predicate Logic Representation

Let:

- $RiverBank(b) = b$ is a riverbank
- $River(r) = r$ is a river
- $Storm(s) = s$ is a storm
- $Flooded(b) =$ the bank flooded

- $\text{LocatedBy}(b,r)$ = bank b is beside river r
- $\text{SavedByAction}(b,a)$ = bank b was saved by action a
- $\text{QuickAction}(a)$ = a was a quick action

Predicate representation:

$\text{RiverBank}(b)$

$\text{River}(r)$

$\text{LocatedBy}(b,r)$

$\text{Storm}(s) \text{ Flooded}(b)$

$\text{After}(\text{Flooded}(b), s)$

$\text{QuickAction}(a)$

$\text{SavedByAction}(b,a)$

A compact logical representation is:

$\text{RiverBank}(b) \wedge \text{LocatedBy}(b,r) \wedge \text{River}(r)$

$\wedge \text{Storm}(s) \wedge \text{Flooded}(b)$

$\wedge \text{After}(\text{Flooded}(b), s)$

$\wedge \text{QuickAction}(a) \wedge \text{SavedByAction}(b,a)$

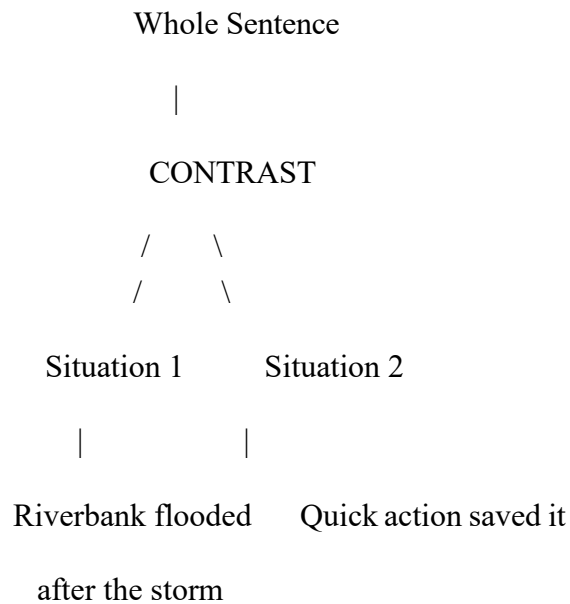
c) Target Sentence / Paraphrase

The riverbank near the river flooded after the storm, but quick action saved it.

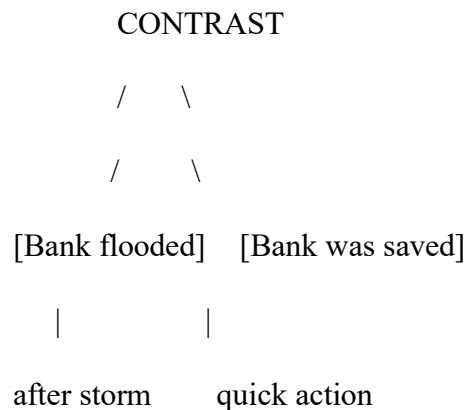
This preserves the original meaning and removes the ambiguity of “bank.”

d) RST-Style Discourse Tree

The relation between the two clauses is **Contrast**.



Or:



e) Advantages of Constraint-Based Approach

A constraint-based approach has several advantages over a purely statistical machine translation approach:

1. Better ambiguity resolution
Contextual constraints identify “bank” as riverbank.
2. Preserves semantic meaning
The important relationships between the bank, river, storm and action are maintained.
3. Maintains discourse structure
The contrast relationship between the two clauses is preserved.
4. Reduces incorrect translations

A statistical system might select “financial bank” based on frequency, while constraints use the surrounding context.

5. Improves coherence

The translated sentence remains logically connected and semantically consistent.

Rubrics for Calculation

Criteria		Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	
Problem Understanding		25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	
Constraint Analysis		25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	
Solution Development		25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	
Application of Concepts		15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	
Justification		10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	
Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %

1							
2							
3							

Faculty In-charge		Course Coordinator		Program Director			
Signature: _____		Signature: _____		Signature: _____			
Date: _____		Date: _____		Date: _____			